

USB Device Monitor

An Android app that monitors USB device attach/detach events for the **Zebra RFD40 RFID Sled** and displays live status on a modern Material UI.

Features

- **Live status indicator** — large green circle (☑ RFD40 Awake) when the device is connected; gray circle (☐ Sleep Mode) when disconnected
- **Device details** — shows Device Count, Vendor ID (VID 1504), Product ID (PID), and device path on connect
- **USB transport & power monitoring** — decodes the active USB mode (Power-Only / Charging vs. File Transfer / Debug) from USB_STATE and power connect/disconnect broadcasts
- **Interface-change counters** — running totals of USB **attach**, **detach**, and combined **total** events for the current session
- **Automatic RFID reader control** — on RFD40 attach the app initializes the Zebra RFID SDK and connects the reader; on detach it disconnects and cleans up
- **Event log** — timestamped scrollable log of all USB attach/detach, transport, power, and RFID reader events
- **Wake hint** — prompts user to “*Hold RFD40 trigger to wake up RFID Reader*” when in sleep mode
- **Toast notifications** on each attach/detach event

Screenshots

Awake (Connected)	Sleep Mode (Disconnected)
Green circle, ☑ icon, VID/PID populated	Gray circle, ☐ icon, wake instruction shown

Requirements

- Android **API 34+** (Android 14)
- Zebra RFD40 RFID Sled (VID 1504, PID 5889)
- USB Host mode enabled on the Android device
- **Zebra RFID API3 SDK .aar** placed in app/libs/ (proprietary — see app/libs/README.md)

Project Structure

```
app/src/main/
├── java/com/zebra/rfid/usb/
│   ├── MainActivity.java          # USB BroadcastReceiver + UI logic
│   └── RFIDHandler.java          # Zebra RFID SDK lifecycle (init/connect/disconnect)
├── libs/                          # Zebra RFID API3 SDK .aar goes here
├── res/
│   ├── layout/activity_main.xml  # Material card-based UI
│   └── drawable/
│       ├── shape_circle_awake.xml # Green status circle
│       └── shape_circle_sleep.xml # Gray status circle
└── build_run.sh                  # Clean → Build → Install → Launch script
```

Build & Run

One-command script (requires connected device)

```
./build_run.sh
```

This script: 1. Checks for a connected ADB device/emulator 2. Runs `./gradlew clean` 3. Runs `./gradlew assembleDebug` 4. Installs the APK via `adb install -r` 5. Launches MainActivity 6. Streams MYUSB logcat output

Manual

```
./gradlew clean assembleDebug
adb install -r app/build/outputs/apk/debug/app-debug.apk
adb shell am start -n com.zebra.rfid.usb/.MainActivity
```

USB Event Handling

The app registers **two** runtime BroadcastReceivers.

mUsbReceiver — host attach/detach:

Intent Action	Meaning
USB_DEVICE_ATTACHED	RFD40 plugged in — green circle, populate VID/PID
USB_DEVICE_DETACHED	RFD40 unplugged — gray circle, clear device info

mBatteryReceiver — transport mode + power state:

Intent Action	Meaning
USB_STATE	Decode transport mode via <code>updateUsbModeFromIntent()</code> (connected, configured, mtp, ptp, mass_storage, adb, rndis)
ACTION_POWER_CONNECTED	Cable connected — classify as <i>Power-Only (Charging)</i> or <i>File Transfer / Debug</i>
ACTION_POWER_DISCONNECTED	Cable removed — reset decoded USB state

Mode determination: explicit transport function extras (mtp / ptp / mass_storage / adb / rndis) take priority; if none are present, a connected && configured USB is inferred as a data-capable (file-transfer/debug) link.

See USB_ATTACH_DETACH_DESIGN.md for the full design document with code snippets.

Interface-Change Counters

The app maintains session counters for USB interface changes, shown in the **USB INTERFACE CHANGES** card:

Counter	Increments on	Color
Attach (mAttachCount)	ACTION_USB_DEVICE_ATTACHED	green
Detach (mDetachCount)	ACTION_USB_DEVICE_DETACHED	red
Total	attach + detach (derived)	blue

Each increment is also appended to the event log (Attach count = N / Detach count = N). Counters are session-scoped — they start at 0 on launch and are not persisted across activity recreation.

RFID SDK Integration

When the Zebra RFD40 sled is attached/detached, MainActivity drives the Zebra RFID API3 SDK through a dedicated RFIDHandler:

USB Event	RFIDHandler call	Action
USB_DEVICE_ATTACHED (VID 1504)	onUsbAttached()	Initialize SDK □ enumerate readers □ connect □ configure event listeners
USB_DEVICE_DETACHED	onUsbDetached()	Remove listeners □ disconnect reader □ release reader state
onDestroy()	onDestroy()	Dispose SDK + shut down background executor

Reader status (connecting, connected, disconnected, tag reads, errors) is routed back into the event log with a [RFID] prefix via a StatusListener callback. SDK work runs on a single-threaded background executor with volatile guards (initializationInProgress, connectionInProgress, resumeRequested, readersAttached) to avoid duplicate init/connect races.

SDK requirement: the proprietary Zebra RFID API3 .aar is **not** bundled in this repository. Drop it into app/libs/ (see app/libs/README.md). app/build.gradle pulls it in via implementation fileTree(dir: 'libs', include: ['*.aar', '*.jar']). Until present, the com.zebra.rfid.api3.* symbols in RFIDHandler.java will not resolve and the project will not build.

See USB_ATTACH_DETACH_DESIGN.md §10 for the full integration design.

Key Log Tags

```
adb logcat -s MYUSB:D MYUSB_STATE:D
D Device List Size: 1
D Device VID: 1504
D Device PID: 5889
D USB action: android.hardware.usb.action.USB_DEVICE_DETACHED
D USB device detached
D USB action: android.hardware.usb.action.USB_DEVICE_ATTACHED
D ACTION_USB_DEVICE_ATTACHED
D --> Vid =1504
```

USB transport/power events are logged under the MYUSB_STATE tag:

```
D USB_STATE[broadcast] connected=true, configured=true, mtp=false, ptp=false, mass_storage=false,
D ACTION_POWER_CONNECTED mode=File Transfer / Debug, connected=true, configured=true, fileTransfe
D ACTION_POWER_DISCONNECTED
```

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